

Who Built the Service Economy?

Immo Schott

Federal Reserve Board

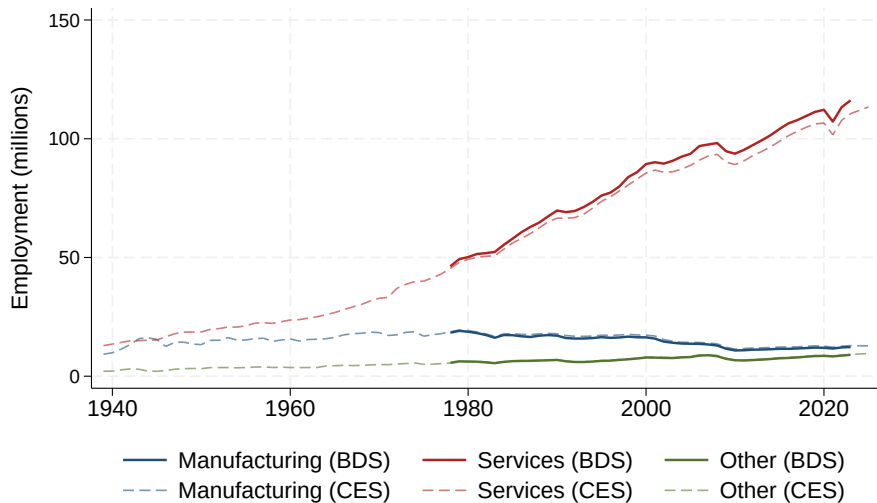
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Sectoral transformation

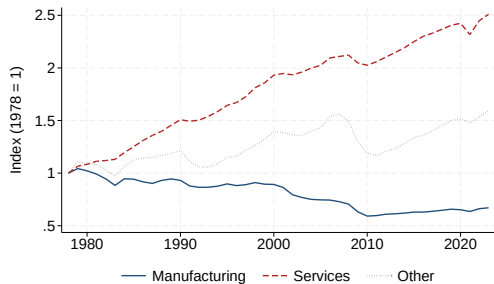


Our questions:

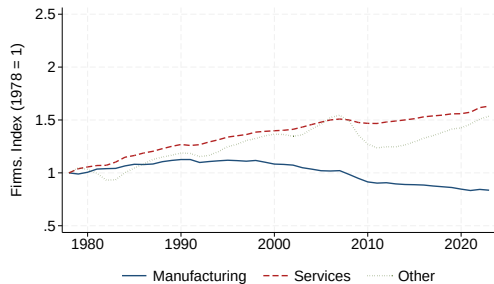
- ▶ How did this large reallocation of labor come about?
- ▶ Was it driven by entry of new service-sector firms?
- ▶ Or by the decline and exit of manufacturing incumbents?
- ▶ Did structural transformation primarily reflect ...
- ▶ ... the rise of small, young service firms?
- ▶ ... or the expansion of large, established incumbents?

We take a firm-dynamics perspective: which firms grew, and how?

Service employment up 2.5x, firm count only 1.6x

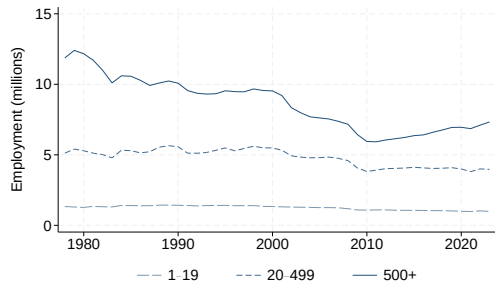


Employment (1978 = 1)

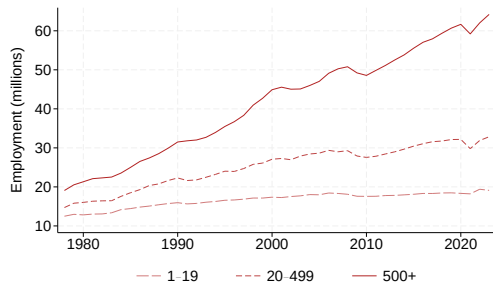


Number of firms (1978 = 1)

Employment share of large service firms increased

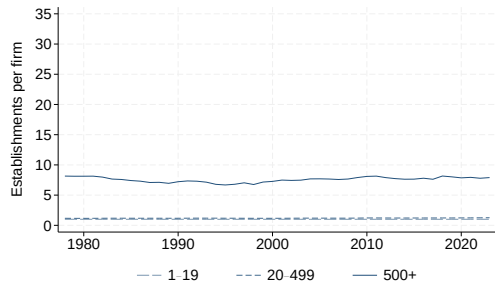


Manufacturing

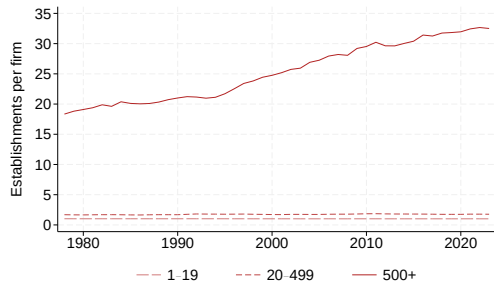


Services

Large service firms grew by adding establishments



Manufacturing



Services

Empirical takeaway: incumbents built the service economy

- ▶ Service employment rose 2.5-fold, while the number of service firms rose only 1.6-fold.
- ▶ Employment growth was concentrated in large service firms.
- ▶ Large firms expanded their establishment networks, while employment per establishment remained broadly stable.
- ▶ Incumbents account for 93% of service-employment growth.
 - ▶ 41 p.p. arise directly from opening additional establishments; the remainder is net growth at continuing locations.

But...

- ▶ "...isn't this just health care?". No. Service Subsector Levels Service Employment Gains
- ▶ "...did service establishments grow in all service industries?" Yes. By Industry
- ▶ "...new establishments, or acquired ones?" New establishments. Births
- ▶ "...did establishments get bigger?" Not in services. Emp. per Establishment
- ▶ "...what about entry and exit?" Falling until mid 2010s. Employment Share Firm Rates
- ▶ "...what about firm age?" Similar picture. Firm Age
- ▶ "...did you formally decompose ΔE ?" Yes. Decomposition

Related literature

Structural transformation and multisector growth

Herrendorf, Rogerson & Valentinyi (2014), Ngai & Pissarides (2007), Herrendorf et al. (2015), Kongsamut, Rebelo & Xie (2001), Buera & Kaboski (2012), Comin, Lashkari & Mestieri (2021)

- **This paper:** adds incumbent establishment decisions to standard ST models

Spatial expansion and multi-establishment firms

Hsieh & Rossi-Hansberg (2023), Oberfield et al. (2024), Kleinman (2025), Cao, Hyatt, Mukoyama & Sager (2022), Lashkari et al. (2024), Brynjolfsson et al. (2025), Argente et al. (2025), Gumpert et al. (2022)

- **This paper:** quantifies how lower expansion costs reallocate employment across sectors

Firm dynamics

Hopenhayn (1992), Hopenhayn & Rogerson (1993), Luttmer (2007), Decker et al. (2014), Hopenhayn et al. (2022), Karahan et al. (2024), Peters & Walsh (2026), Hopenhayn et al. (2025), Klette & Kortum (2004), Bernard et al. (2010), Aghion et al. (2023), Xi (2023)

- **This paper:** incumbent expansion dominates entry in service employment growth

Model

What we do

We extend Hopenhayn & Rogerson (1993) to two sectors:

- two final-goods sectors $j \in \{M, S\}$, linked by non-homothetic CES demand
- multi-establishment firms with convex **establishment-replication costs**

Two aggregate driving forces (1979 vs. 2023):

- **higher** manufacturing productivity $a_M \rightarrow$ drives sectoral reallocation
- **lower** replication-cost convexity $\eta \rightarrow$ incumbents scale via establishments

What we find:

- the standard productivity channel generates sectoral reallocation but leaves firm policies unchanged
- cheaper replication delivers large-firm, multi-establishment growth
- ... and contributes to ST itself: **1/3** of the model-generated reallocation

Households

Representative household:

- supplies labor $\bar{L}$ inelastically
- income $I_t = W_t \bar{L} + \Pi_t$ (wages plus rebated firm profits)
- maximizes $\mathbb{E}_0 \sum_t \beta^t \ln C_t$ over a CES composite of M & S

Non-homothetic demand:

- a manufacturing subsistence level $\bar{c}_M$ is purchased first
- only the remaining (supernumerary) spending is split across M & S
- as income rises, the manufacturing share of spending falls
- needed so demand shifts toward services, beyond relative prices

Demand equations

Firms and establishments

Each establishment uses a decreasing-returns technology:

$$q = a_j z l^{\alpha_j}, \quad \alpha_j \in (0, 1).$$

A firm runs $m \geq 1$ identical establishments, subject to replication costs:

$$G(m) = c_{mj} \frac{m^{1+\eta_j} - 1}{1 + \eta_j}.$$

Firm flow profit in sector j is:

$$\Pi_j(z) = \max_{m \geq 1, l \geq 0} m[p_j a_j z l^{\alpha_j} - Wl] - P \cdot G(m) - P \cdot f_j.$$

Free entry, endogenous & exogenous exit

Entry & exit

Stationary equilibrium

Extensive margin of firm growth

Establishment-level optimization: decreasing returns

$$l^*(z) = \left(\frac{p_j \alpha_j a_j z}{W} \right)^{\frac{1}{1-\alpha_j}}, \quad \pi_v(z) \propto z^{\frac{1}{1-\alpha_j}}.$$

Optimal number of establishments per firm

$$m^*(z) = \max \left\{ 1, \left(\frac{\pi_v(z)}{P_{Cmj}} \right)^{1/\eta_j} \right\}.$$

- ▶ Intensive margin: larger establishments; extensive margin: more establishments.
- ▶ Only firms with $\pi_v(z) > P_{Cmj}$ replicate; convexity η_j sets how fast.

Effective returns to scale

Effective firm-level returns to scale exceed establishment-level returns:

$$L_j(z) = m_j^*(z)l_j^*(z) \propto z^{\frac{1}{1-\alpha_{\text{eff},j}}}, \quad \alpha_{\text{eff},j} = \frac{1 + \eta_j \alpha_j}{1 + \eta_j} \in (\alpha_j, 1).$$

- $\eta_j \rightarrow \infty$: $\alpha_{\text{eff},j} \rightarrow \alpha_j$ (single establishment)
- $\eta_j \rightarrow 0$: $\alpha_{\text{eff},j} \rightarrow 1$ (constant returns)

Establishment growth and productivity

Lower η_j tilts establishment growth toward high-productivity firms:

$$\log m_j^*(z) = \kappa_j + \frac{1}{\eta_j(1 - \alpha_j)} \log z, \quad m_j^*(z) > 1,$$

- lower η_j raises the coefficient on $\log z$ and makes establishment scope more elastic to productivity
- because the left side is $\log m_j^*$, high- z firms expand proportionally more

More analytical results in the paper:

- lower η_j generates the high- z expansion and raises effective firm-level returns to scale
- sectoral productivity changes relative prices but leaves firm policies and the size distribution unchanged

η proposition

Productivity proposition

Where does a lower η come from? Three recent models

Hsieh & Rossi-Hansberg (2023): units are **markets**

- a firm pays a fixed cost to raise productivity in all of its markets
- firms with more markets gain more, adopt more intensively, and enter still more markets

Brynjolfsson et al. (2025): units are **establishments**

- IT lets a firm run all of its branches the way its best branch runs
- the more branches a firm has, the more it gains, so large chains expand

Argente et al. (2025): units are **products**

- when demand rises, it pays to standardize know-how shared by all products
- productive firms standardize the most, so new products are cheap for them

Isomorphic to a lower η :

- large firms gain more from each change because they have more units
- that is what a lower η does: m^* rises mostly at the top of the distribution

Quantitative results

Parameterization

Assigned outside the model

- β, ε at standard values; exit rates from BDS, p_S/p_M from BEA

Household

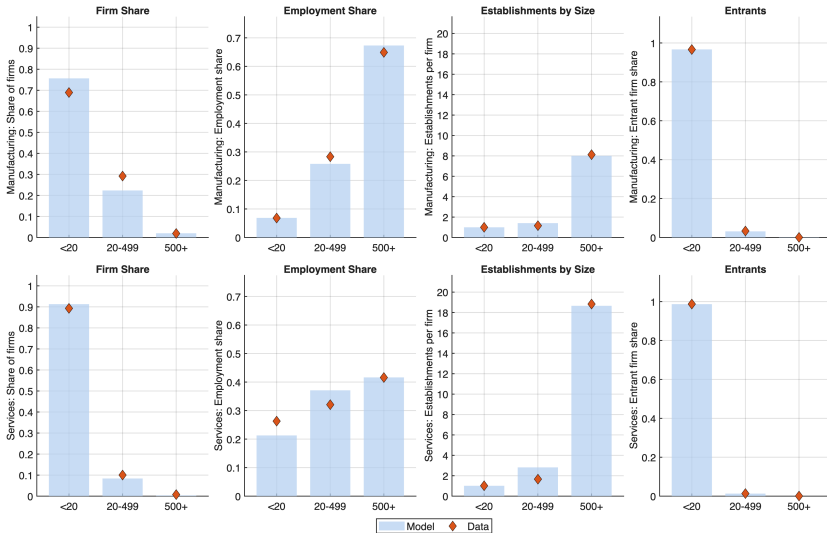
- $\bar{c}_M$ and μ : goods expenditure share and manufacturing employment share

Firms: sector by sector, 1979 BDS moments

Table

- productivity and entrant parameters: firm-size distribution, young-firm growth
- α_j : average establishment size
- $(c_{m,j}, \eta_j)$: establishments per firm, overall and large relative to small

Initial (1979) steady-state fit



Untargeted: fit by firm age

Quantitative experiments

Goal: capture the two forces behind structural transformation

1. manufacturing productivity growth: $a_M \uparrow$
2. organizational change, cheaper replication: $\eta \downarrow$, proportionally in both sectors

From initial 1979 steady state to a new 2023 steady state

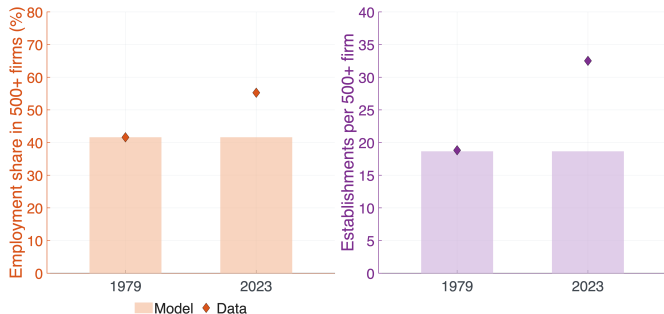
- a_M matches the 1979–2023 change in the relative price of services
- η matches the rise in establishments per firm, large vs. small service firms
- all other primitives fixed; employment share of large service firms untargeted

What we do:

- **Experiment 1:** $a_M \uparrow$ only, the standard structural-transformation mechanism
- **Experiment 2:** $a_M \uparrow$ and $\eta \downarrow$ together, our main experiment

Experiment 1: Raise manufacturing productivity (a_M) only

Raise a_M ($\times 1.55$) to match the manufacturing vs. service relative-price change



- ✓ generates structural transformation: manufacturing employment share 28.0% $\rightarrow$ 19.9% (44% of decline)
- ✗ firm-size distribution does not move: $a_M \uparrow$ is offset by $p_M \downarrow$
- ✗ no shift to large service firms (41.6% vs. 55.3% in data), no establishment growth

Experiment 2: a_M rises and replication gets cheaper $\eta \downarrow$

$a_M \uparrow$ accompanied by lower replication costs:

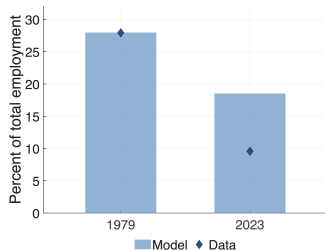
- lower η makes it less costly to add establishments
- η falls by the same proportion in both sectors

Some suggestive evidence [Details](#)

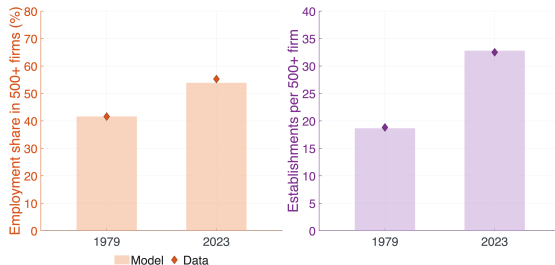
- IT capital per worker $\uparrow$, office & admin. share $\downarrow$
- the fall is linked to more establishments per firm
- ...and especially pronounced in *large* firms

Experiment 2: Model fit

$a_M \times 1.57$, $\eta \times 0.80$ (both sectors): matches the micro facts



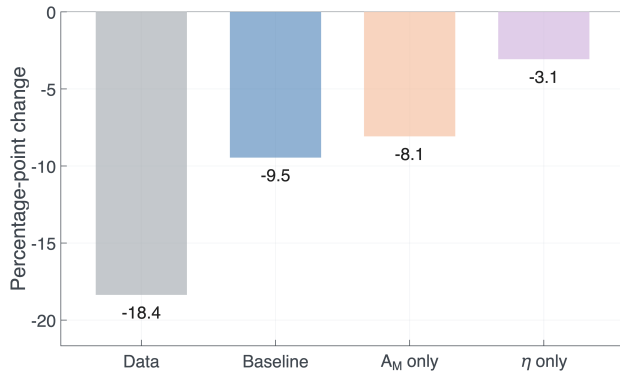
Manufacturing employment
share



Large (500+) service firms

- ✓ generates structural transformation: manufacturing employment share 28.0% → 18.5% (52% of decline)
- ✓ establishments per large service firm 18.7 → 32.8 (targeted)
- ✓ employment share of large service firms 41.6% → 53.9% (untargeted; data 55.3%)

Each force alone: Decline in manufacturing employment share



- a_M only: 44% of the decline, but no shift toward large service firms
- η only: 1/3 of the structural transformation explained by the model
- the two do not sum to 52%: each triggers its own price adjustment

Adding labor-force growth

Main experiment with labor supply scaled by 1.88 (BDS, 1979–2023)

	Main experiment	+ labor growth	Data
Manufacturing employment share, 2023 (%)	18.5	14.8	9.6
Share of decline explained (%)	51.5	71.8	—
Employment share of large service firms (%)	53.9	53.9	55.3
Establishments per large service firm	32.8	32.8	32.5

Labor-force growth matters for the aggregate, not for the firm-level facts

- the model now accounts for 72% of the manufacturing employment decline

Next steps: Implications and robustness

Implications (in progress)

- concentration: model explains much of the rising 500+ employment share
- one establishment per firm: how much of the aggregate shift is lost?

Robustness

- endogenous exit: results essentially unchanged; does the entry rate fall?
- non-homothetic CES demand: results change little
- convex entry costs instead of free entry: to do

Conclusion

Structural transformation through the lens of firm dynamics

- the reallocation toward services is not driven by entry
- large incumbent service firms expand, mainly along the extensive margin

Implications for modeling

- treating sectors as representative firms misses a key margin of adjustment
- multi-establishment firms: model explains half of the M share decline
- lower replication costs alone deliver 1/3 of the ST the model explains

Who built the service economy?

- existing service firms that expanded establishments across markets

Appendix

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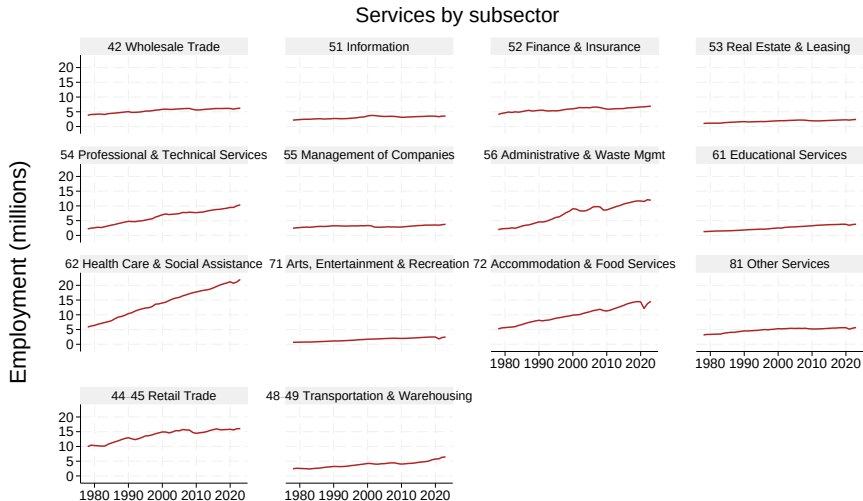
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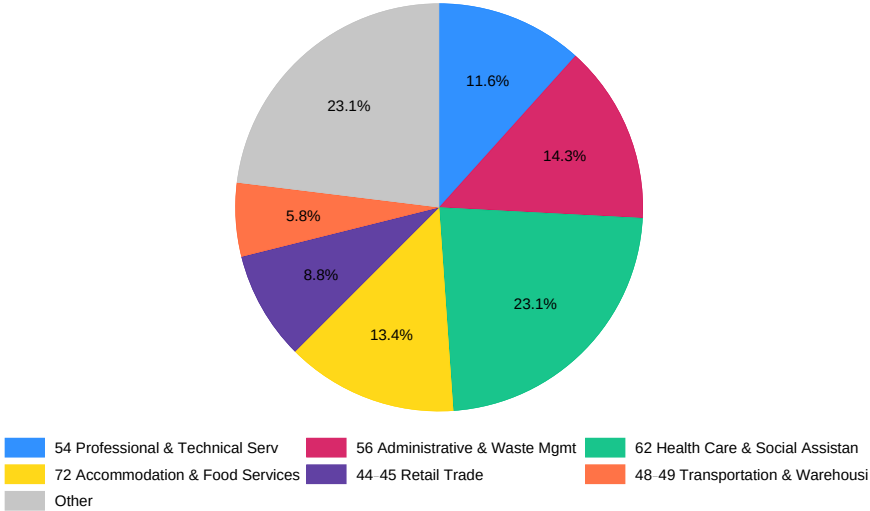
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Service Employment by Subsector

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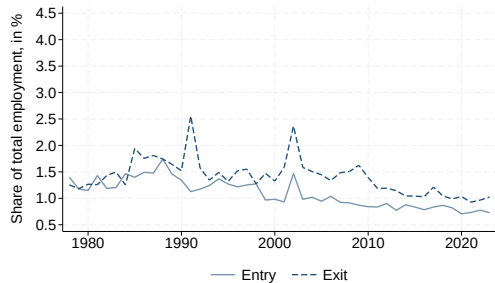
Service Employment Gains by Subsector

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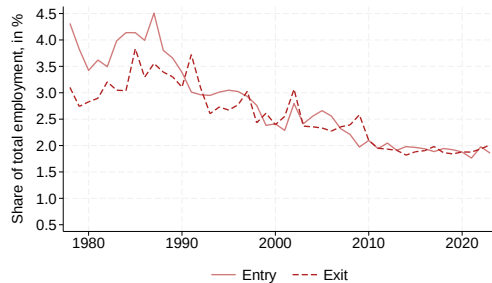


Entry and Exit Employment Shares

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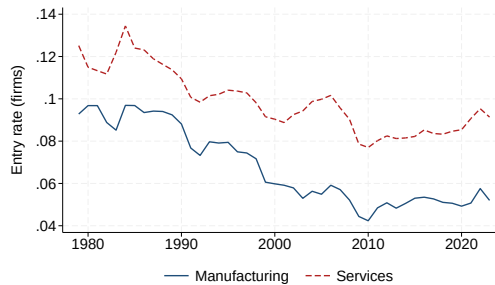
a) Manufacturing



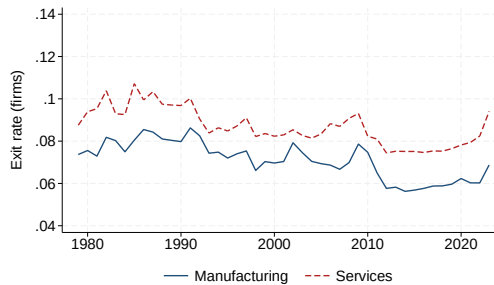
b) Services

Entry and Exit Rates

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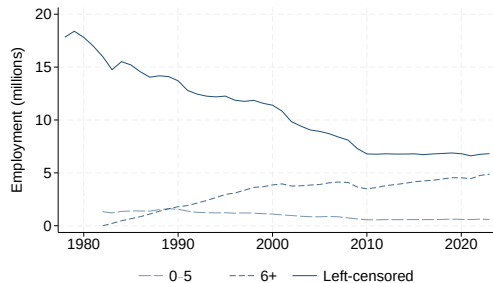
a) Entry rates



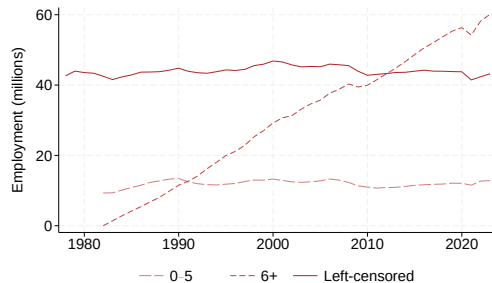
b) Exit rates

Employment by Firm Age

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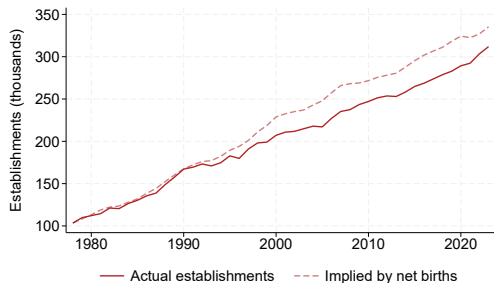


a) Manufacturing

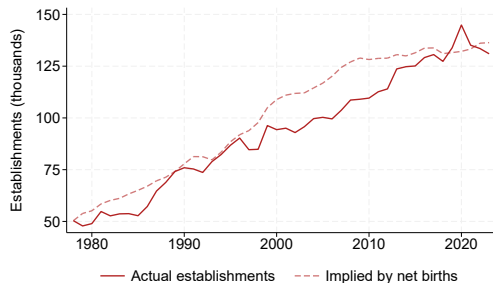


b) Services

New establishments, not acquisitions

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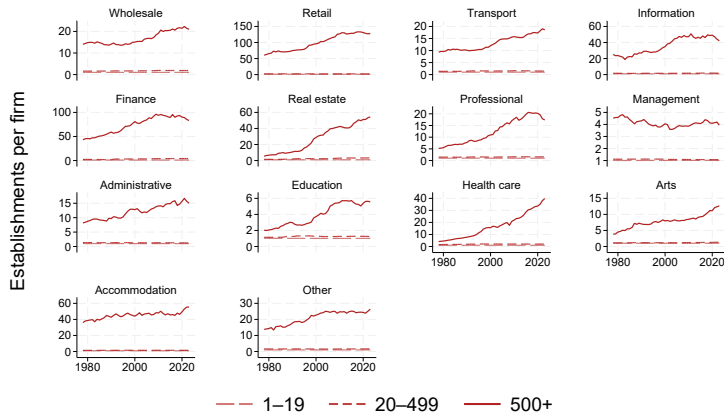
a) Firms with 500–2,499 employees



b) Firms with 2,500–4,999 employees

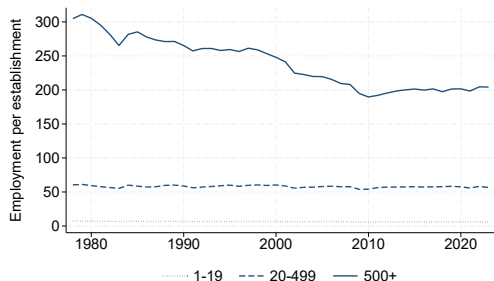
The dashed line adds cumulative net establishment births to the 1978 stock. Its proximity to the observed stock implies that acquisitions play a limited role.

Establishments per firm across service industries

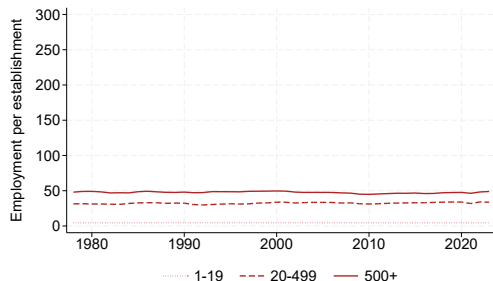
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Firms with 500 or more employees add establishments in nearly every two-digit service industry.

Employment per establishment by firm size

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a) Manufacturing



b) Services

Service establishment size stays stable, so large service firms grow through additional establishments.

Household demand

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Households maximize:

$$U = \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \ln(C_t), \quad I_t = W_t \bar{L} + \Pi_t.$$

Manufacturing has a subsistence component:

$$C_{Mt} = \bar{c}_M + \tilde{C}_{Mt}, \quad C_{St} = \tilde{C}_{St}.$$

Supernumerary consumption is aggregated as:

$$C_t = \left[\mu^{\frac{1}{\varepsilon}} \tilde{C}_{Mt}^{\frac{\varepsilon-1}{\varepsilon}} + (1-\mu)^{\frac{1}{\varepsilon}} C_{St}^{\frac{\varepsilon-1}{\varepsilon}} \right]^{\frac{\varepsilon}{\varepsilon-1}},$$

subject to:

$$P_{Mt} \bar{c}_M + P_t C_t = I_t, \quad P_t = \left[\mu P_{Mt}^{1-\varepsilon} + (1-\mu) P_{St}^{1-\varepsilon} \right]^{\frac{1}{1-\varepsilon}}.$$

$$V_j(z) = \Pi_j(z) + \beta(1 - p_{\text{exit}}) \max\{0, \mathbb{E}[V_j(z') \mid z]\}.$$

- ▶ Conditional on surviving, the firm chooses whether to continue.
- ▶ Endogenous exit occurs when $\mathbb{E}[V_j(z') \mid z] < 0$.
- ▶ Entrants pay $c_{e,j}$, draw $z \sim \pi_{e,j}$, and enter until:

$$\mathbb{E}_{\pi_{e,j}}[V_j(z)] = Pc_{e,j}.$$

Stationary equilibrium

Stationary Equilibrium

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- ▶ Households allocate consumption between manufacturing and service goods.
- ▶ Firm policies solve the establishment, replication, continuation, and exit problems.
- ▶ Free entry holds in each sector:

$$\mathbb{E}_{\pi_{e,j}}[V_j(z)] = P c_{e,j}$$

- ▶ Supernumerary expenditure shares are:

$$\phi_j = \omega_j \left(\frac{p_j}{P} \right)^{1-\varepsilon}, \quad \omega_M = \mu, \quad \omega_S = 1 - \mu$$

- ▶ Goods markets clear, with C_M including $\bar{c}_M$:

$$M_j \int Y_j(z) d\mu_j(z) = C_j + \phi_j B$$

- ▶ Labor market clears:

$$\sum_{j \in \{M, S\}} M_j \int l_j(z) m_j(z) d\mu_j(z) = \bar{L}$$

Proposition: Sectoral productivity

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Analytical variant: homothetic demand and labor-denominated costs.

Invariance of firm policies to sectoral productivity

For each sector $j \in \{M, S\}$, define $x_j \equiv p_j a_j$. Then:

- (i) $l_j^*(z)$, $m_j^*(z)$, profits, the value function, and exit depend on (p_j, a_j) only through x_j ;
- (ii) free entry uniquely determines x_j . If $a'_j = \lambda a_j$, then $p'_j = p_j/\lambda$;
- (iii) the stationary firm-size distribution, normalized by the mass of firms, is invariant to a_j .

Replication-cost proposition

Proposition: Replication costs and effective returns

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Conditional on prices, let $\mathcal{I}_j \equiv \{z : \pi_{v,j}(z) > c_{m,j}\}$.

For $z \notin \mathcal{I}_j$, $m_j^*(z) = 1$ and is independent of η_j . For $z \in \mathcal{I}_j$,

$$\frac{\partial \log m_j^*(z)}{\partial \eta_j} < 0, \quad \frac{\partial^2 \log m_j^*(z)}{\partial \eta_j \partial \log z} < 0.$$

Thus, a reduction in η_j raises the number of establishments, with a larger proportional effect at higher productivity.

For firms in the interior region,

$$L_j(z) \propto z^{\frac{1}{1-\alpha_{\text{eff},j}}}, \quad \alpha_{\text{eff},j} = \frac{1 + \eta_j \alpha_j}{1 + \eta_j} \in (\alpha_j, 1), \quad \frac{\partial \alpha_{\text{eff},j}}{\partial \eta_j} < 0.$$

Productivity proposition

$$\begin{aligned}\Delta E_{s,t} = & \underbrace{\left(JC_{s,t}^{\text{birth}, f=0} - \text{FirmDeathEmp}_{s,t} \right)}_{\text{Net firm entry}} + \underbrace{\left(JC_{s,t}^{\text{cont}} - JD_{s,t}^{\text{cont}} \right)}_{\text{Net within-establishment expansion of incumbents}} \\ & + \underbrace{\left(JC_{s,t}^{\text{birth}, f \geq 1} - \left(JD_{s,t}^{\text{death}} - \text{FirmDeathEmp}_{s,t} \right) \right)}_{\text{Net establishment expansion within incumbent firms}}.\end{aligned}$$

$JC^{\text{birth}, f=0}$: job creation by startups, JC^{cont} , JD^{cont} : job creation/destruction by continuing establishments, $JC^{\text{birth}, f \geq 1}$: new establishments within incumbent firms.

Net entry explains only 7% of service employment growth

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Component	Δ employment (m)	Share (%)
<i>Manufacturing, 1978–2023</i>		
Net firm entry	−2.07	37.7
Net expansion of continuing establishments	−0.60	11.0
Net establishment expansion within incumbents	−2.82	51.3
Total	−5.48	100.0
<i>Services, 1978–2023</i>		
Net firm entry	4.88	6.8
Net expansion of continuing establishments	37.60	52.6
Net establishment expansion within incumbents	29.00	40.6
Total	71.48	100.0

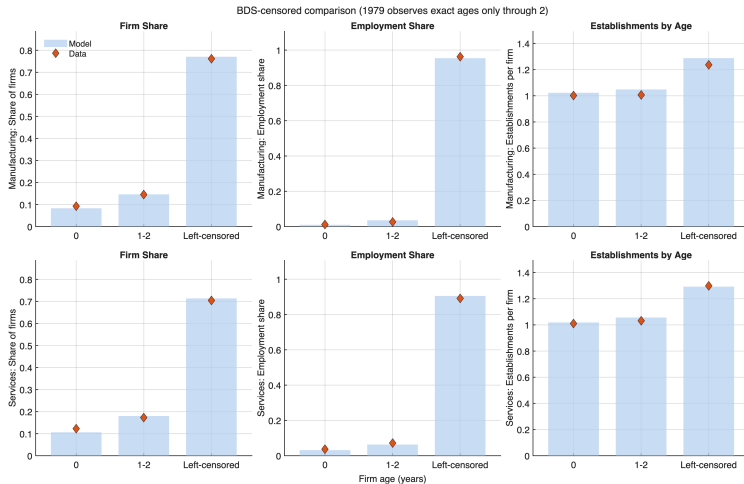
Gross entry and exit flows are large, but mostly offset each other.

Calibration to the 1979 economy

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Parameter	Description	Value	Target	Data	Model
<i>Parameters assigned outside the model</i>					
β	Discount factor	0.96			
ε	Substitution elasticity	0.40	Desmet and Rossi-Hansberg (2014)		
a_M, a_S	Sectoral TFP	1.00	Normalization		
δ_M	Exogenous exit rate	0.08	Average firm entry and exit rate		
δ_S	Exogenous exit rate	0.11	Average firm entry and exit rate		
p_S/p_M	Relative price of services	0.64	BEA value-added price indexes		
<i>Household</i>					
$\bar{c}_M$	Manufacturing subsistence	0.32	Goods-share slope wrt log PCE per capita	-0.23	-0.23
μ	CES preference weight	0.09	Manufacturing employment share	0.28	0.28
<i>Manufacturing</i>					
μ_M^z	Mean log productivity	1.81	Average firm size	73.21	74.16
σ_M	Productivity volatility	0.44	Employment share, firms with 500+	0.65	0.67
$\mu_{e,M}^z$	Mean log entrant productivity	0.27	Average entrant size	9.04	9.03
$\sigma_{e,M}$	Entrant productivity dispersion	0.88	Entrant firm share with 20+	0.03	0.03
ρ_M	Productivity persistence	0.69	Age 1–5 firm share with 20+	0.15	0.15
α_M	Establishment returns to scale	0.53	Average establishment size	61.72	60.21
$c_{m,M}$	Replication-cost level	52.87	Average establishments per firm	1.19	1.23
η_M	Replication-cost convexity	0.87	Estab./firm: 500+ relative to <20	8.10	8.00
$c_{e,M}$	Entry cost	386.56	Free-entry condition	—	—
<i>Services</i>					
μ_S^z	Mean log productivity	0.14	Average firm size	16.43	16.43
σ_S	Productivity volatility	0.18	Employment share, firms with 500+	0.42	0.42
$\mu_{e,S}^z$	Mean log entrant productivity	1.95	Average entrant size	4.90	4.90
$\sigma_{e,S}$	Entrant productivity dispersion	0.24	Entrant firm share with 20+	0.01	0.01
ρ_S	Productivity persistence	0.99	Age 1–5 firm share with 20+	0.06	0.06
α_S	Establishment returns to scale	0.47	Average establishment size	13.46	13.46
$c_{m,S}$	Replication-cost level	14.12	Average establishments per firm	1.22	1.22
η_S	Replication-cost convexity	0.85	Estab./firm: 500+ relative to <20	18.44	18.44
$c_{e,S}$	Entry cost	94.86	Free-entry condition	—	—

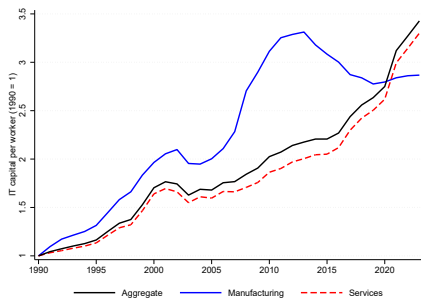
Untargeted fit by firm age

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The BDS age panel starts in 1977. In 1979, exact ages are observed only for age 0 and ages 1–2; firms aged 3+ are pooled in both model and data.

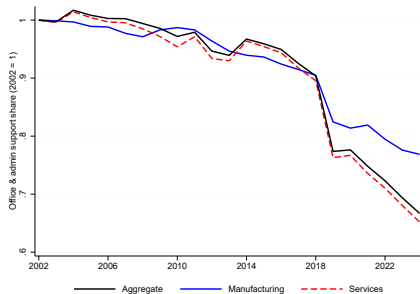
Suggestive evidence: replicability cost has fallen

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a)

IT capital per worker rises



b)

Office and admin share falls

- ▶ +1 SD IT-intensity growth: +0.088 log establishments per firm for large firms relative to small firms.
- ▶ +1 SD fall in office/admin share: same +0.088 relative increase.
- ▶ No robust link to firm count or employment per establishment.

Summary of quantitative experiments

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	A_M, η	A_M only	η only	$+\bar{L}$	Data
A_M (factor, 1979 = 1)	1.57	1.55	1.00	1.57	
Replication-cost scaling	0.80	1.00	0.70	0.80	
η_S	0.67	0.85	0.59	0.67	
Labor supply $\bar{L}$	1.00	1.00	1.00	1.88	
Manufacturing employment share (%)	18.52	19.89	24.90	14.79	9.57
Share of decline explained (%)	51.5	44.0	16.7	71.8	
Large-firm share of service employment (%)	53.9	41.6	68.9	53.9	55.3
Establishments per service firm, 500+	32.8	18.7	32.6	32.8	32.5
Average establishments per service firm	1.28	1.22	1.33	1.28	1.37